54.

SOLUTION :

Given :

$T_i = 298$ K

$T_f = 298.45$ K

$k = 2.5$ kJ K^{-1}

$m = 3.5g$

$M_m = 28$

heat evolved = $k \Delta T$

$= k (T_f - T_i)$

$= 2.5$ kJ K^{-1} $(298.45-298)K$

$= 1.125$kJ

$\Delta H_c = \frac{1.125}{3.5} \times 28$ kJ mol^{-1}

$\Delta H_c = 9$ kJ mol^{-1}

55.

SOLUTION :

Given :

$T_{sys} = 77^0C = (77 + 273) = 350$ K

$T_{surr} = 33^0$ C $= (33 + 273) = 306$ K

$q = 245$ J

$\Delta S_{sys} = \frac{q}{T_{sys}} = \frac{-245}{350} = -0.7$ JK^{-1}

$\Delta S_{surr} = \frac{q}{T_{sys}} = \frac{+245}{306} = +0.8$ JK^{-1}

$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$

$\Delta S_{univ} = -0.7$ $JK^{-1} + 0.8$ JK^{-1}

$\Delta S_{univ} = 0.1$ JK^{-1}

56.

SOLUTION :

Given :

$n = 1$mole

$P = 4.1$ atm

$V = 2$ Lit

$T = ?$

$q = 3710$ J

$\Delta S = \frac{q}{T}$

$\Delta S = \frac{q}{\left(\frac{PV}{nR}\right)}$

$\Delta S = \frac{nRq}{PV}$

$$\Delta S = \frac{1 \times 0.082 \text{ lit atm } K^{-1} \times 3710 \text{ J}}{4.1 \text{ atm} \times 2 \text{ lit}}$$

$$\Delta S = \frac{1 \times 0.082 \text{ lit atm } K^{-1} \times 3710 \text{ J}}{4.1 \text{ atm} \times 2 \text{ lit}}$$

$\Delta S = 37.10$ JK^{-1}

57.

SOLUTION:

Given :

$\Delta H_f(NaCl) = 30.4$ kJ $= 30400$ J mol^{-1}

$\Delta S_f(NaCl) = 28.4$ JK^{-1} mol^{-1}

$T_f = ?$

$\Delta S_f = \frac{\Delta H_f}{T_f}$

$T_f = \frac{\Delta H_f}{\Delta S_f}$

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